

NORTHERN UNIVERSITY OF BUSINESS AND TECHNOLOGY KHULNA

Department of CSE

Scientific Calculator Project

Project Title	Scientific Calculator
Course	Software Development LAB
Course Code	CSE 2216
Programming Language	JAVA
Week	Week 7: Scientific Functions Implementation

1. Introduction

Week 7 focuses on implementing the scientific functions of the Scientific Calculator project. According to the project work plan, this phase adds sin, cos, tan, log, and ln functions, along with square, square root, power, and percentage operations. The week also includes π and e constants. These features extend the calculator beyond basic arithmetic operations and make it suitable for common scientific calculations.

2. Objectives

- Add sin, cos, tan, log, and ln functions.
- Implement square, square root, power, and percentage operations.
- Add π and e constants.
- Integrate the scientific functions with the existing calculator interface.
- Display the calculated results correctly.

3. Scientific Functions Implemented

Function	Purpose	Example
sin	Calculates the sine value of an angle.	sin(30)
cos	Calculates the cosine value of an angle.	cos(60)
tan	Calculates the tangent value of an angle.	tan(45)
log	Calculates the common logarithm.	log(100)
ln	Calculates the natural logarithm.	ln(e)
x^2	Calculates the square of a number.	$5^2 = 25$

$\sqrt{}$	Calculates the square root.	$\sqrt{25} = 5$
x^y	Raises one number to the power of another.	$2^3 = 8$
%	Performs percentage calculation.	25%
π	Provides the mathematical constant pi.	π
e	Provides the mathematical constant e.	e

4. Implementation Approach

The scientific operations are integrated into the existing Java Swing calculator. When a user selects a scientific function, the calculator takes the current input, performs the selected operation, and updates the display with the result. The π and e buttons provide their corresponding mathematical constants. The scientific functions are added while keeping the existing calculator workflow and user interface.

5. System Workflow

- Enter a numerical value.
- Select a scientific function or constant.
- Process the selected calculation.
- Display the calculated result.
- Continue with another calculation if required.

6. Example Calculations

Operation	Input	Result
Square	5	25
Square Root	25	5
Power	2^3	8
sin	30°	0.5
cos	60°	0.5
tan	45°	1
log	100	2
ln	e	1

7. Deliverables

- Updated Java Swing Scientific Calculator program.
- Scientific functions: sin, cos, tan, log, and ln.
- Square, square root, power, and percentage operations.
- π and e constants integrated into the calculator.